

Lycopene-rich treatments modify noneosinophilic airway inflammation in asthma: proof of concept.

Antioxidant-rich diets are associated with reduced asthma prevalence. However, direct evidence that altering intake of antioxidant-rich foods affects asthma is lacking. The objective was to investigate changes in asthma and airway inflammation resulting from a low antioxidant diet and subsequent use of lycopene-rich treatments. Asthmatic adults (n=32) consumed a low antioxidant diet for 10 days, then commenced a randomized, cross-over trial involving 3 x 7 day treatment arms (placebo, tomato extract (45 mg lycopene/day) and tomato juice (45 mg lycopene/day)). With consumption of a low antioxidant diet, plasma carotenoid concentrations decreased, Asthma Control Score worsened, %FEV(1) and %FVC decreased and %sputum neutrophils increased. Treatment with both tomato juice and extract reduced airway neutrophil influx. Treatment with tomato extract also reduced sputum neutrophil elastase activity. In conclusion, dietary antioxidant consumption modifies clinical asthma outcomes. Changing dietary antioxidant intake may be contributing to rising asthma prevalence. Lycopene-rich supplements should be further investigated as a therapeutic intervention.